

**Lecture 1**

**An overview of  
Cardiovascular  
system**

*By*

**Staff members of  
Medical Physiology**

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# Objectives

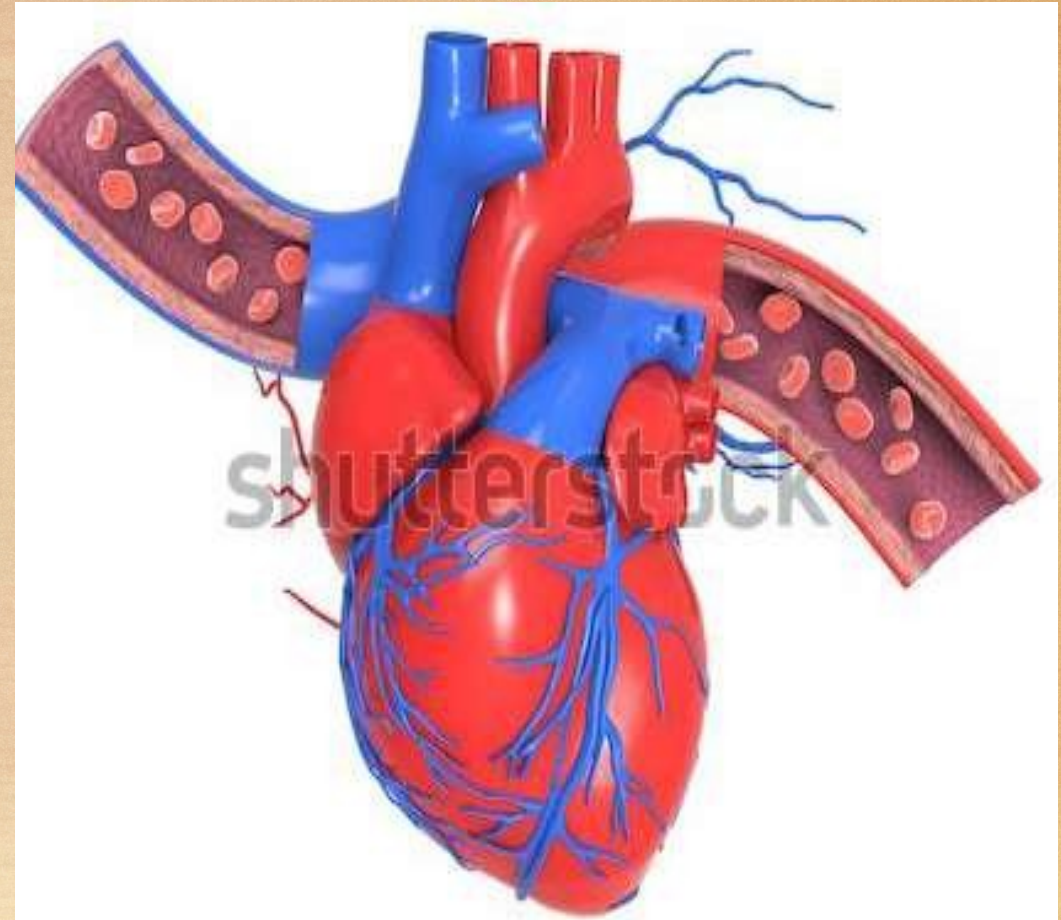
After studying this section, you should be able to:

- ❑ **Identify** the general function of circulatory system
- ❑ **Determine** the division of circulatory system (systemic& pulmonary circulation)
- ❑ **Recognize** the component of CVS and functional parts of systemic circulation
- ❑ **Identify** the physiological anatomy of the heart
- ❑ **Describe** the functional histology of the cardiac muscle

# Introduction to CVS

## Outlines

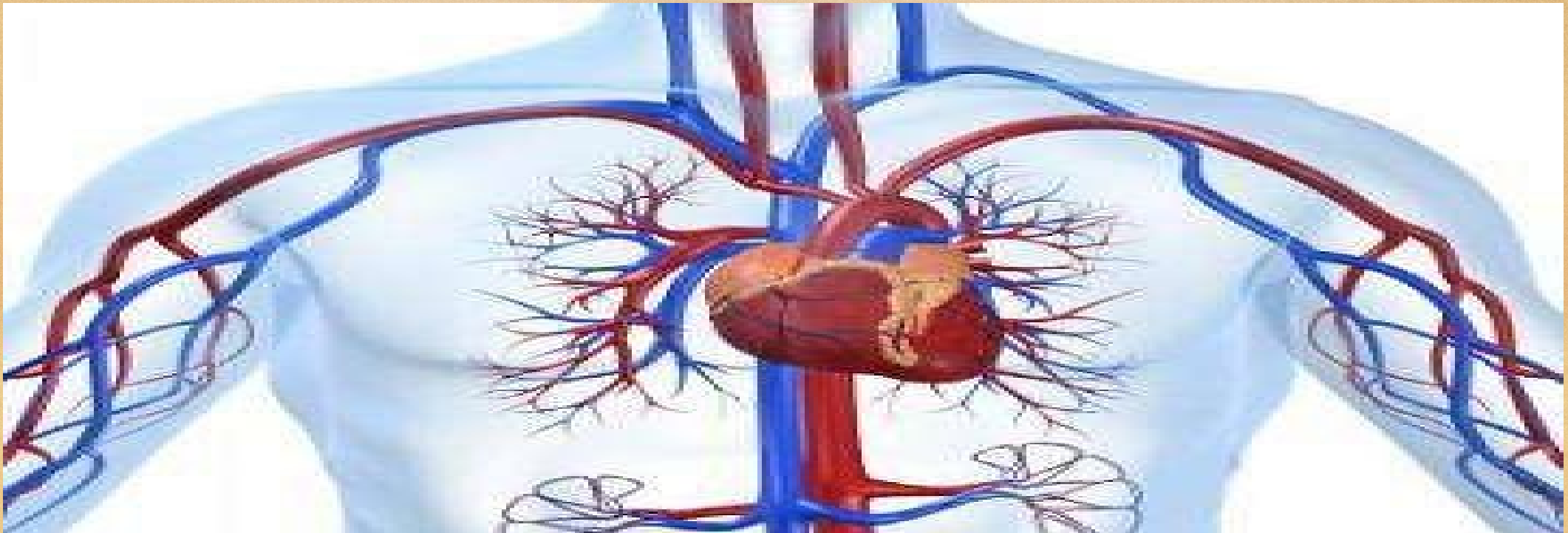
- 1) Functions of the CVS
- 2) Divisions of the Circulatory system
- 3) Components of the heart
- 4) Functional anatomy of the heart
- 5) Blood vessels



# Cardiovascular system (CVS) includes: Heart and blood vessels

- **Heart** pumps blood into the **blood vessels**

- **The blood:** circulates throughout the body to transport nutrients and oxygen to the tissues and removes  $\text{CO}_2$  and waste product from the tissues



The function of the circulatory system:

The main function of circulatory system is to **maintain homeostasis** through:

## **1. Transport function:**

- a. gases** transport of O<sub>2</sub> to the tissues and CO<sub>2</sub> from to the lungs to be excreted to the external environment.
- b. nonvolatile waste products** as urea, uric acid and creatinine to be excreted by the kidneys.
- c.** Transport of the **hormones** secreted by glands to perform its physiological actions.

## **2. Regulation of the body temperature** by controlling the blood flow to the skin.

## **3. Defensive mechanism:** transport of leukocytes and antibodies to the sites of infection

# Divisions of the circulatory system

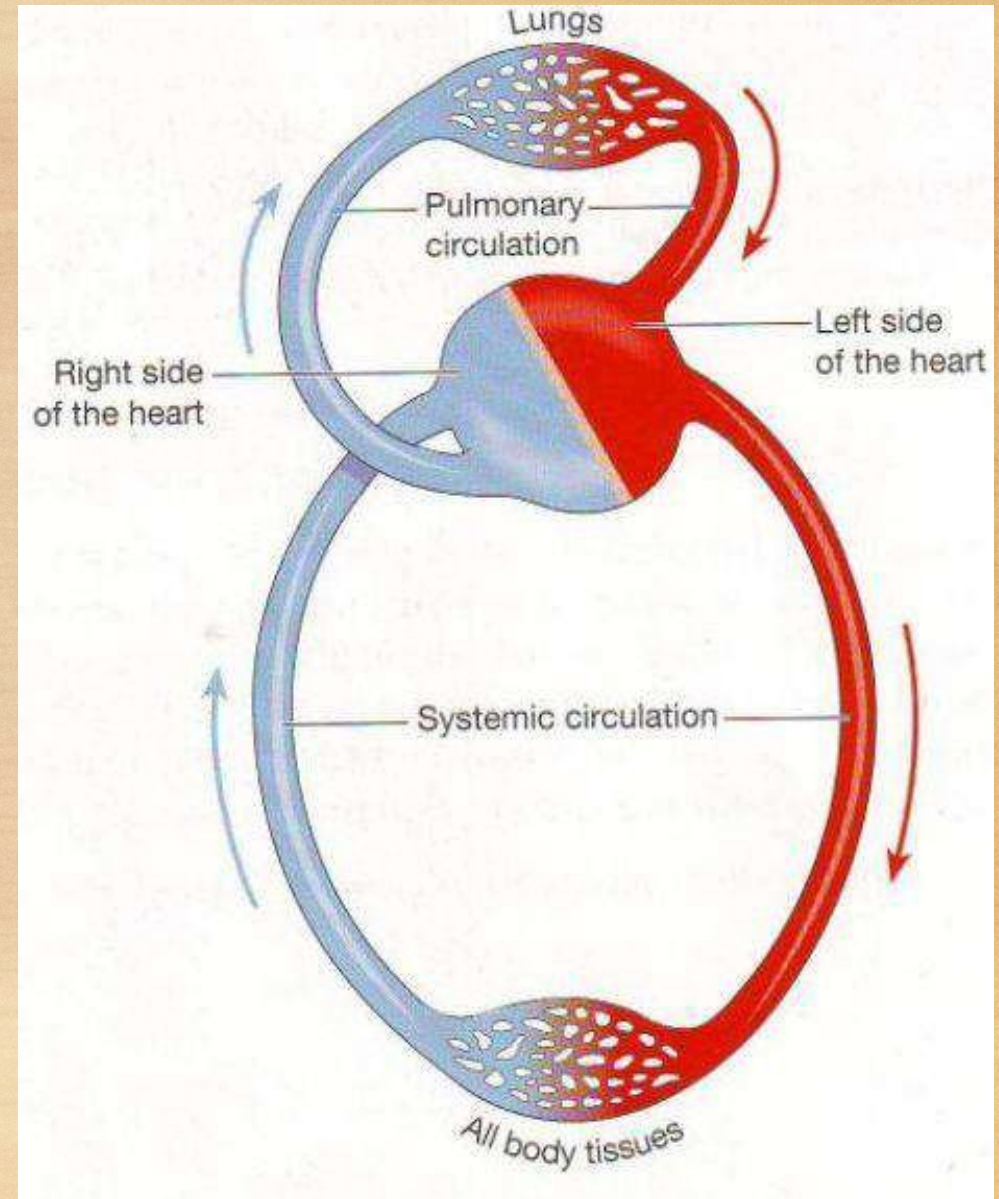
- **Systemic circulation:**

which the blood flows from the left ventricle to the aorta, large arteries, arterioles, and capillaries then to the veins to return back to the right atrium.

- **Pulmonary Circulation:**

In which the blood flows from the right atrium to the right ventricle then through the pulmonary artery to the lungs.

Its aim is gas exchange as  $\text{CO}_2$  is excreted to the outside and  $\text{O}_2$  is transported from the lung alveoli to the blood.

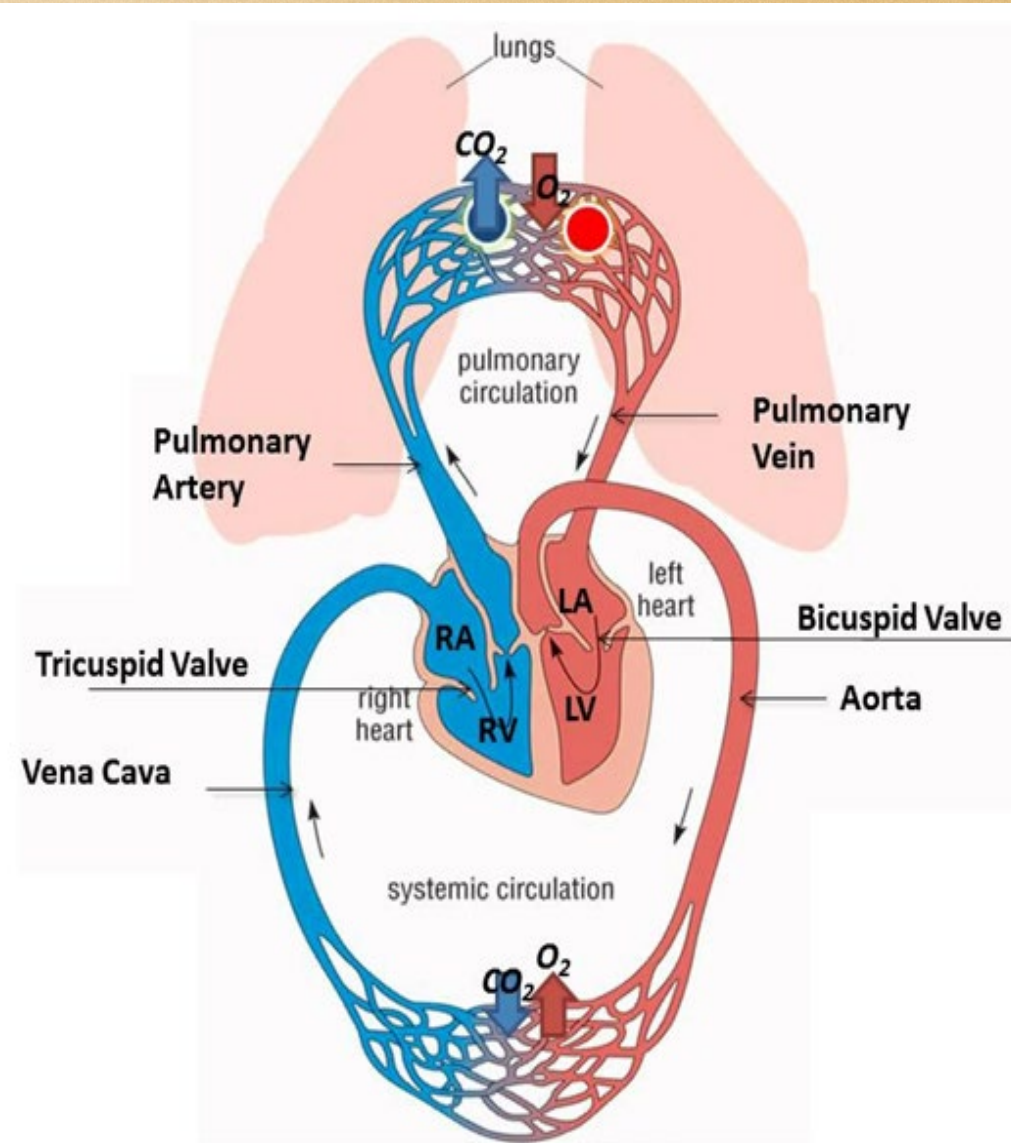


# The heart is formed of 4 chambers

- **2 atria** (Rt and Lt atrium ) separated from each other by interatrial septum
- **2 ventricles** (Rt and Lt ventricle) separated from each other by interventricular septum

-**The Rt. atrium** receives blood from the body through superior and inferior vena cava, and the right ventricle pumps it to the lungs.

-**The left atrium** receives blood from the lungs through the 4 pulmonary veins and the left ventricle pump it to the aorta to reach different organs of the body.



# Myocardium

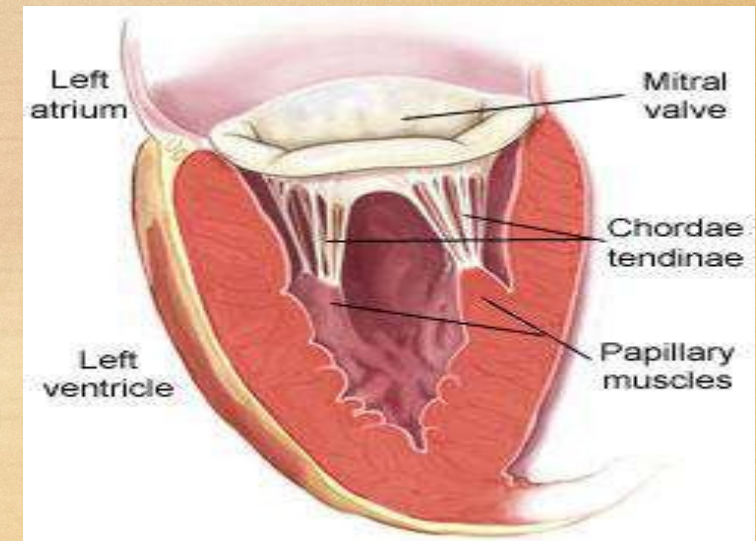
- The atrial muscles: Rt. atrial wall is thinner than the left

## Its functions under resting condition are:

1. It help filling of the ventricles with blood through the atrial contraction.
2. It acts as a blood reservoir of the blood returning to the heart.
3. Regulation of the arterial blood pressure through the cardiovascular reflexes.
4. Secretion of some chemical substances as atrial naturetic peptide(ANP).

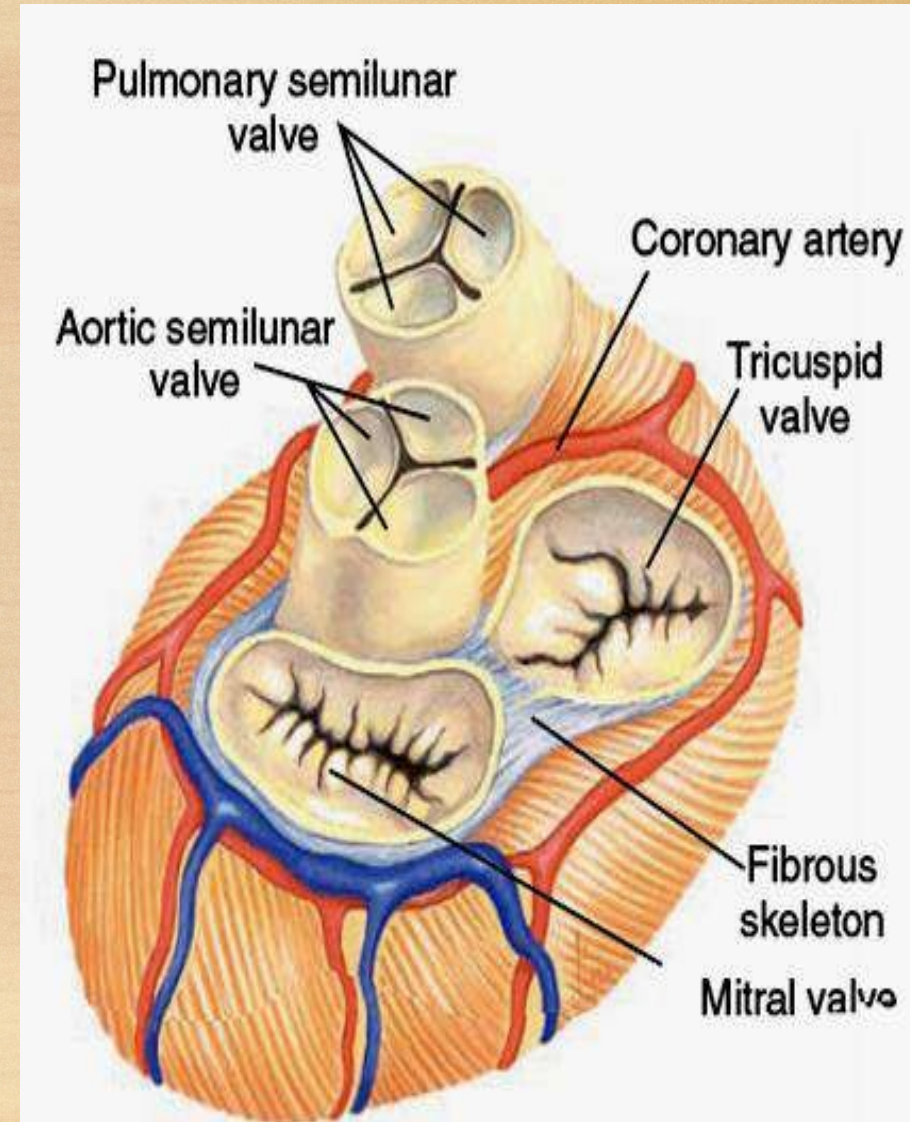
## The ventricular muscle:

The wall of left ventricle is 3 times as thick as the right.



# Fibrous skeleton of the heart

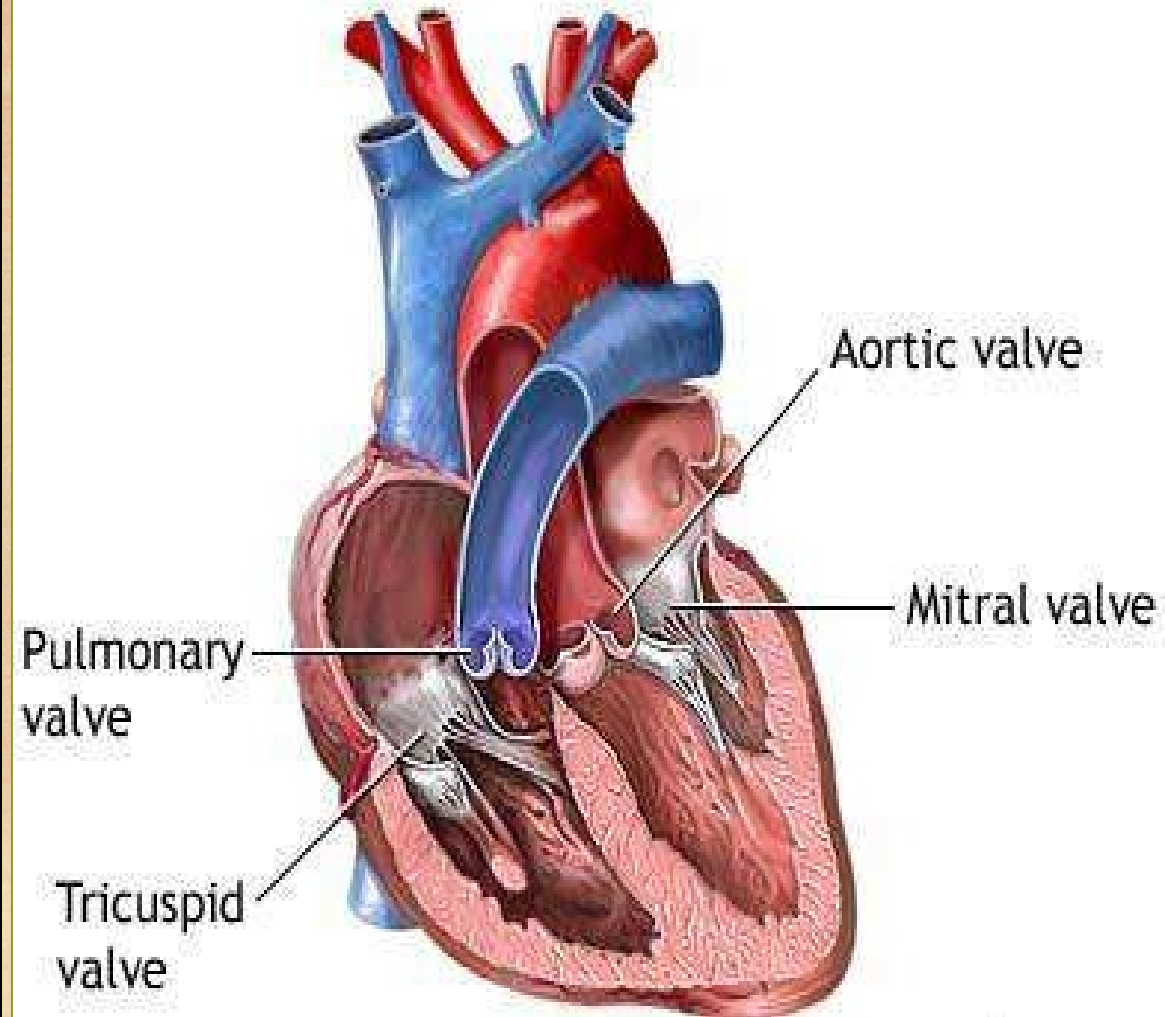
- It separates the **atrial** muscle fibers from **ventricular** muscle fibers.
- It is formed of **4 rings** of dense connective tissue.
- The 4 rings form the openings between atria and ventricles, and between ventricles and major arteries (Pulmonary artery and aorta).
- Each of the four rings has a valve that directs the flow of the blood into one direction.

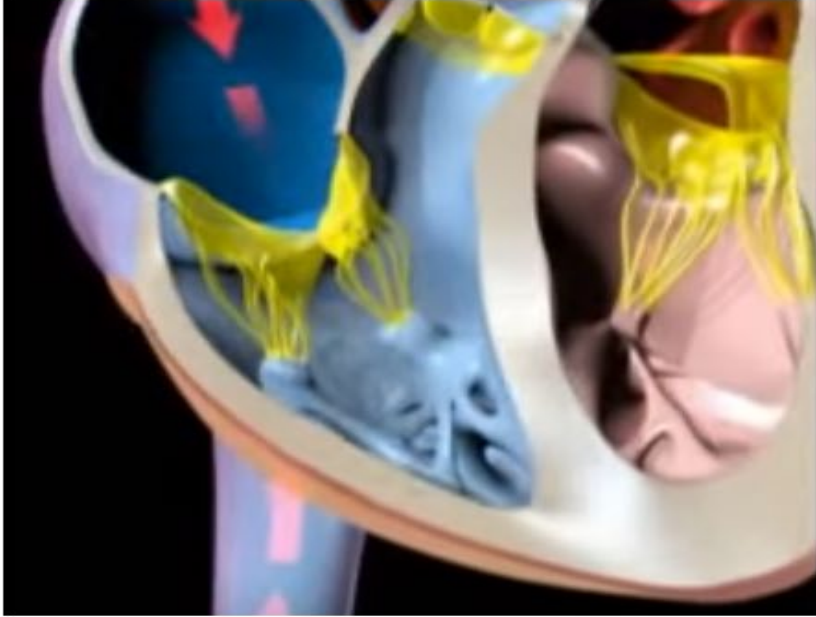


# Cardiac valves

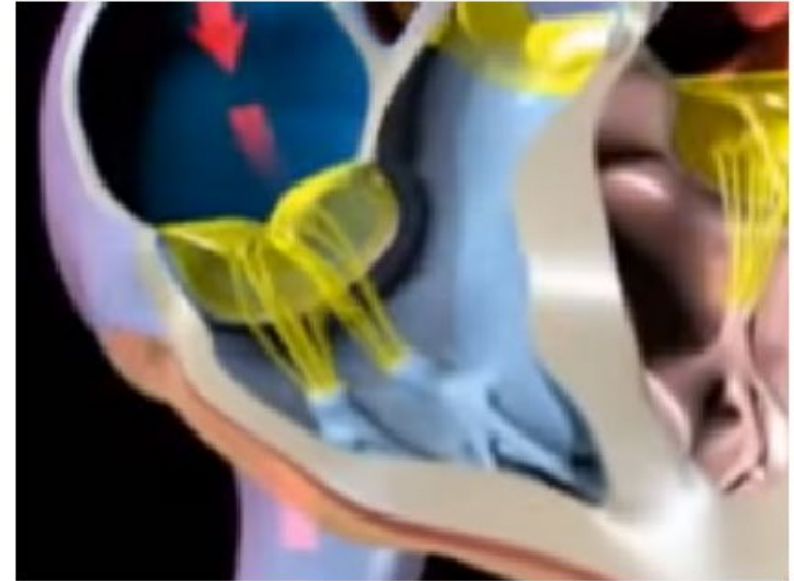
**A] Atrio-ventricular valves (A-V valves):** It separates the atria from the ventricles.

- **Tricuspid valve:** Separate the right atrium from the right ventricle; it is formed of 3 cusps.
- **Mitral valve:** Separate the left atrium from the left ventricle and formed of 2 cusps only.
- The ends of the cusps of A-V valves are attached to the papillary muscles by the chordae tendineae.
- The A-V valves allow the flow of blood in one direction (from atria to the ventricles)
- During ventricular systole, eversion of the A.V valves is prevented by contraction of the papillary muscles.





during relaxation of the ventricle  
opening of cusps of the valve



During ventricular contraction  
Closure of Cusps of the valve

papillary ms prevent eversion of the cusps

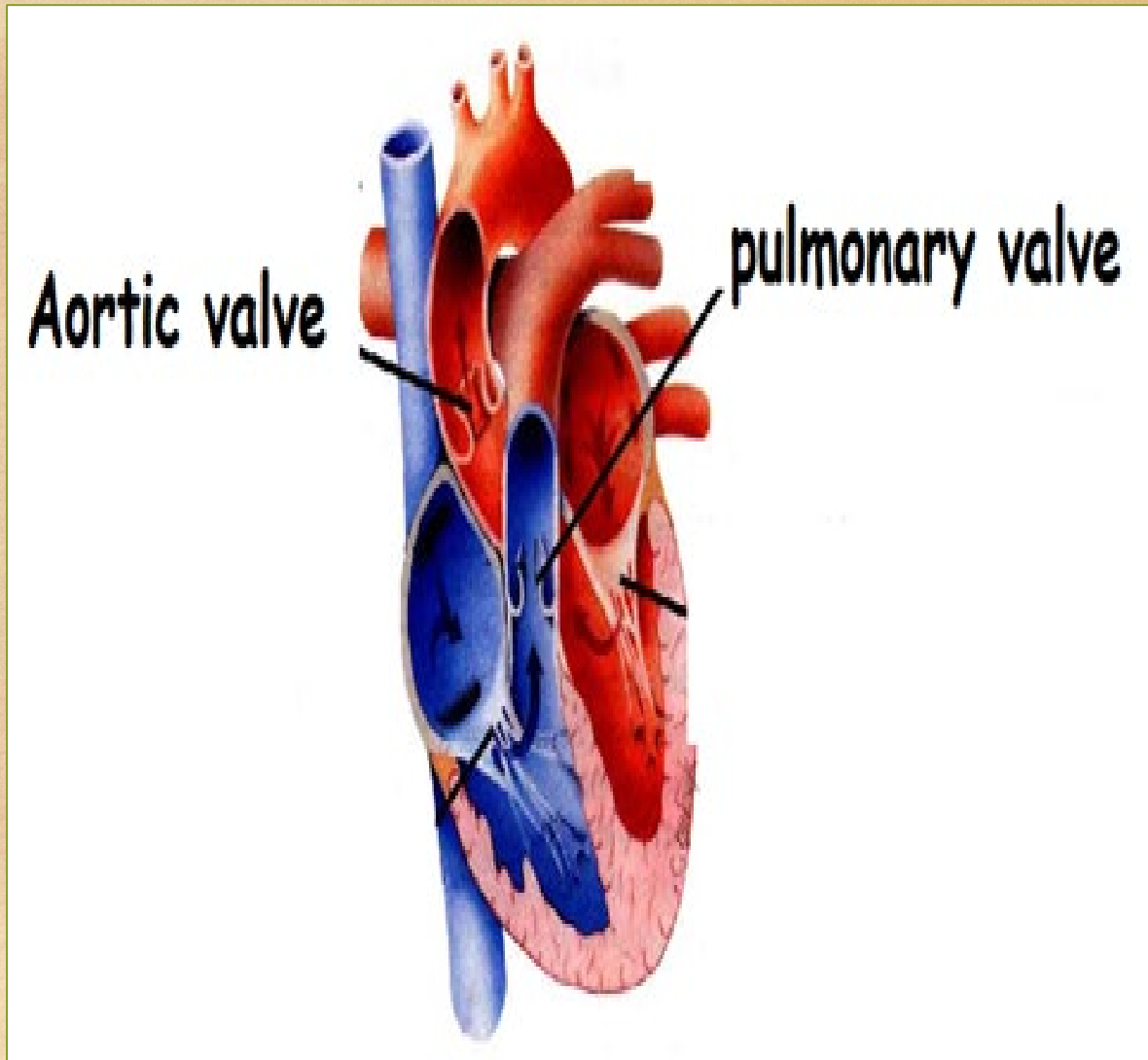
## B] Semilunar valves:

They are the **Aortic** and **pulmonary** valves

- They function as one way valves that allow the flow of blood from the ventricles to the aorta and pulmonary arteries

**N.B:**

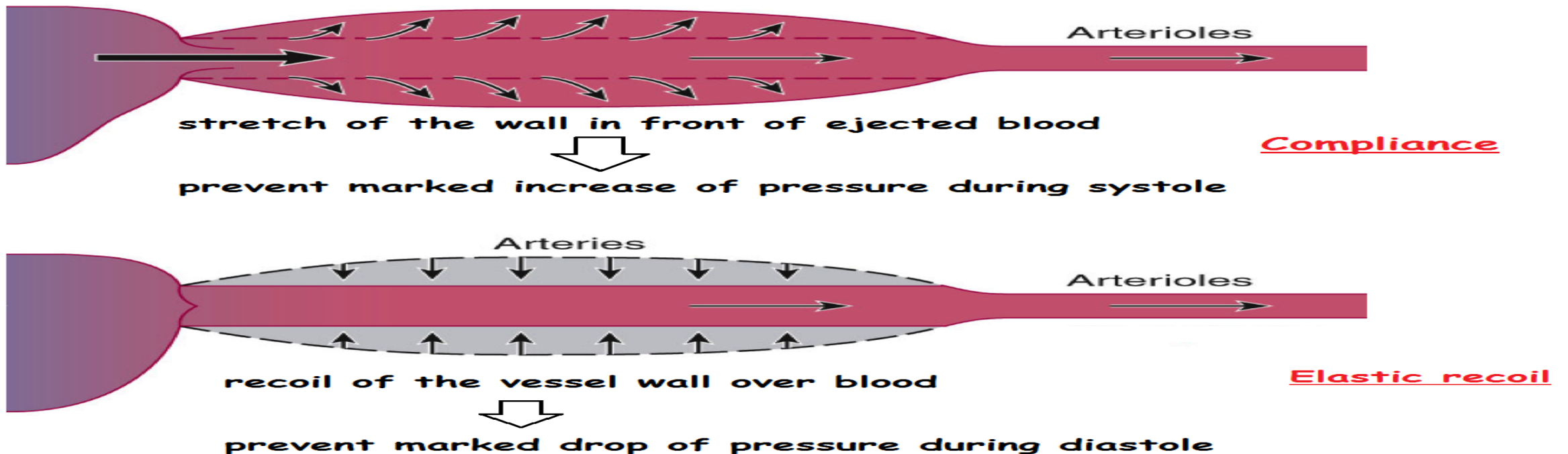
Without the action of the valves blood moves backwards and forwards between the cardiac chambers and major vessels and not pumped to the circulation.



# Blood vessels

## 1- The arteries:

- It contains a lot of elastic fibers in their wall and shows the phenomenon of compliance and elastic recoil.
- **Compliance:** The wall is stretched in front of the ejected blood and prevents marked increase in pressure during systole.
- **Recoil** over the moving blood column prevents marked drop of pressure during diastole.



## 2- Arterioles (resistance vessels):

- They contain a thick layer of smooth muscles in their wall, their diameter is under the control of autonomic nervous system.

- **Sympathetic stimulation** causes V.C increase in arterial blood pressure

- **Parasympathetic stimulation** causes V.D of the arterioles so arterial blood pressure drops.



during rest



V.C sympathetic stim  
++ P.R

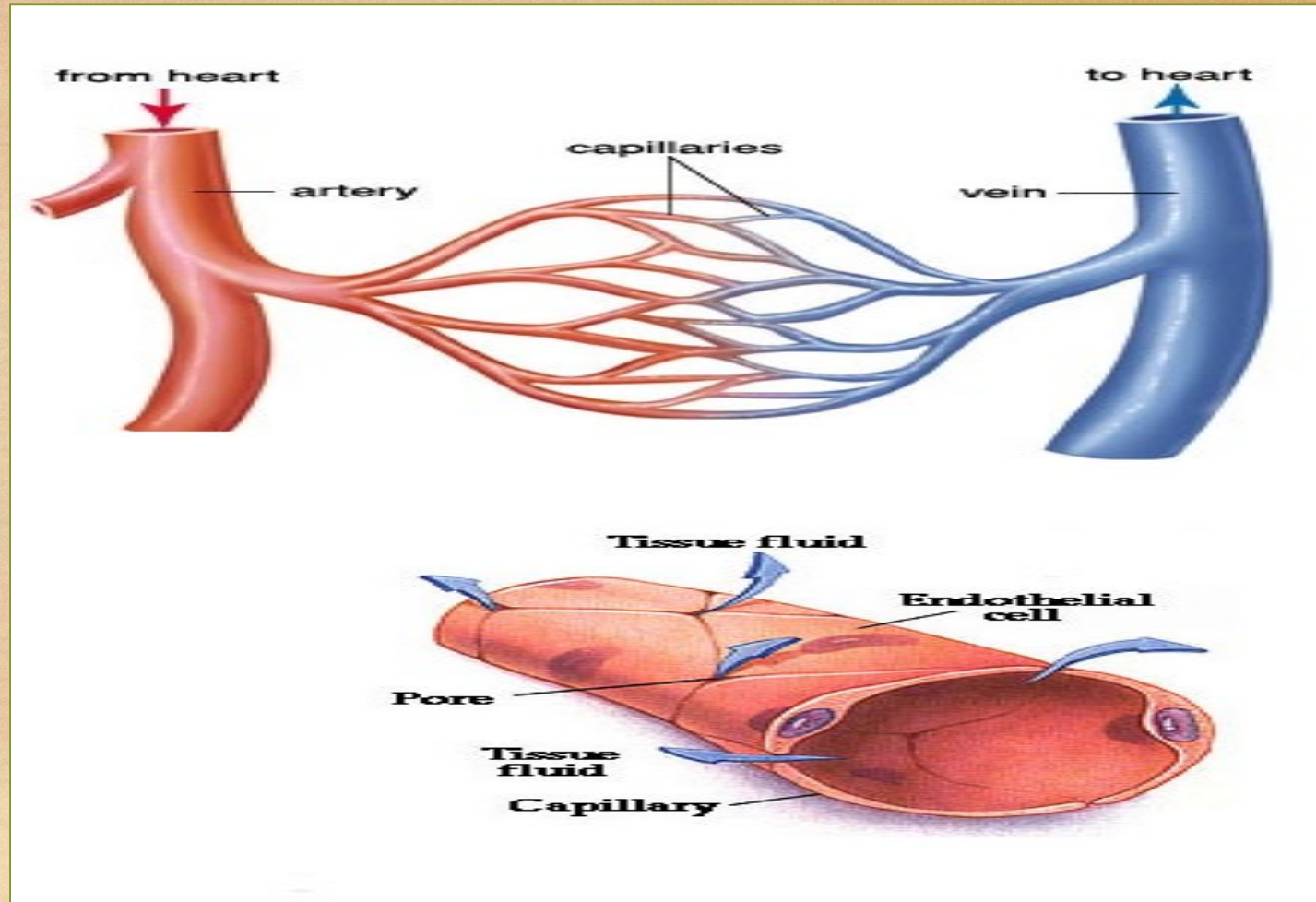


V.D Parasympathetic stim  
- - P.R

### 3-The Capillary ( Exchange vessels ):

The capillary network is the site of contact between the blood and the interstitial cell spaces.

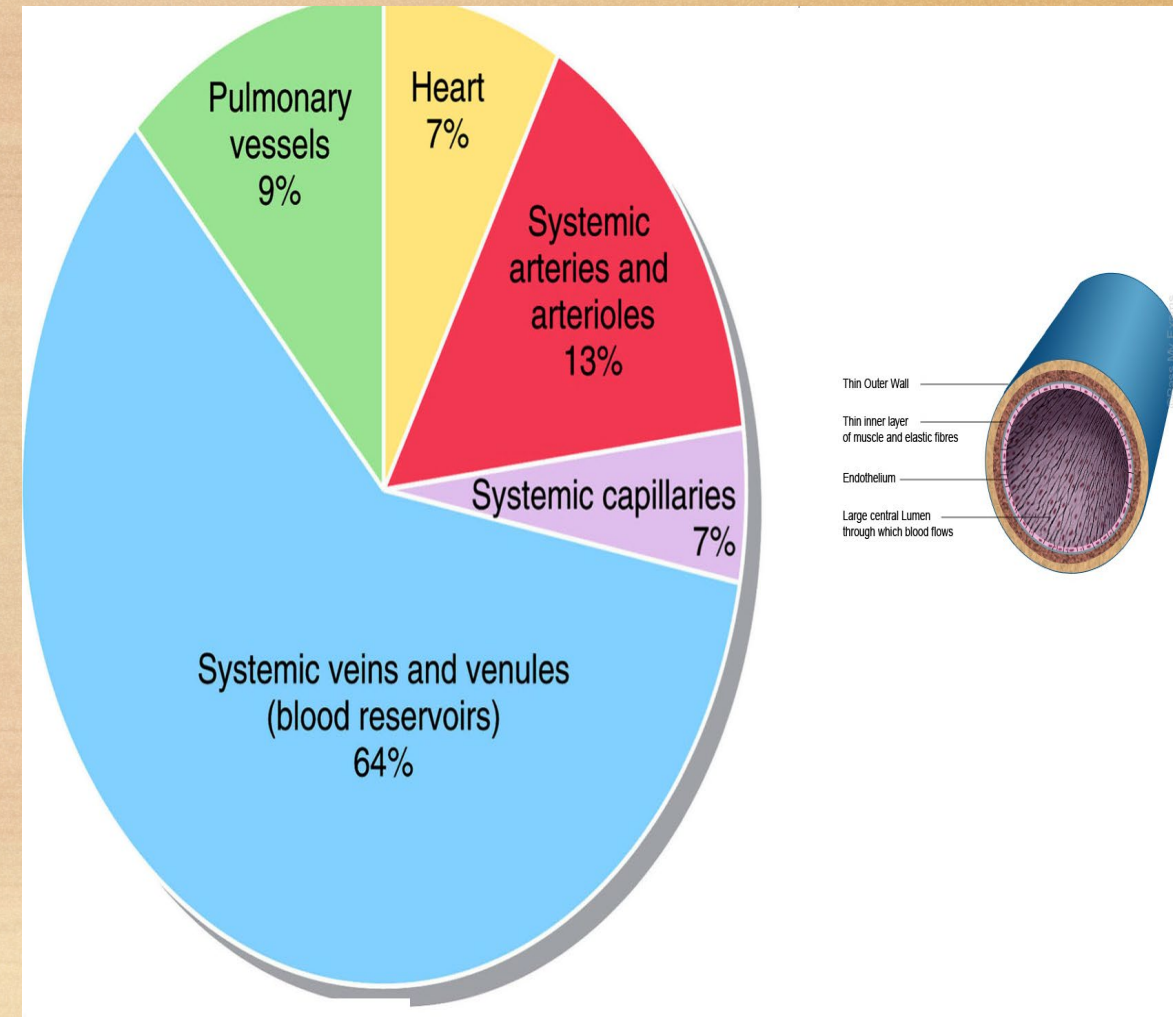
So, it is the only site of CVS through exchange between blood and surrounding tissue occur



## 4- The veins (Capacitance vessels):

They have highly distensible walls. So, they can accommodate a large volume of blood with little change in pressure. The veins can either distend and store or contract and mobilize blood within C.V.S.

**5-Lymphatic Vessels:** Act as a drainage system preventing collection of fluids in the interstitial tissue spaces.



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